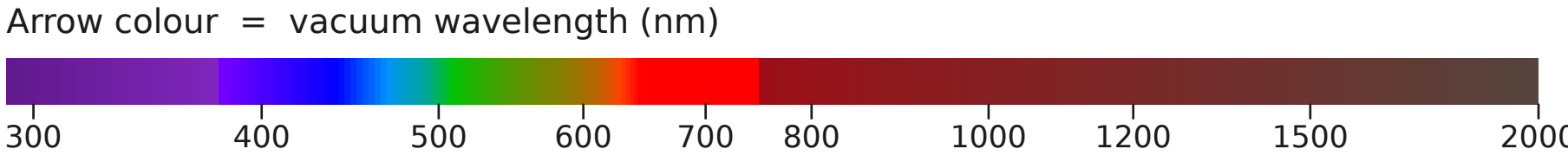


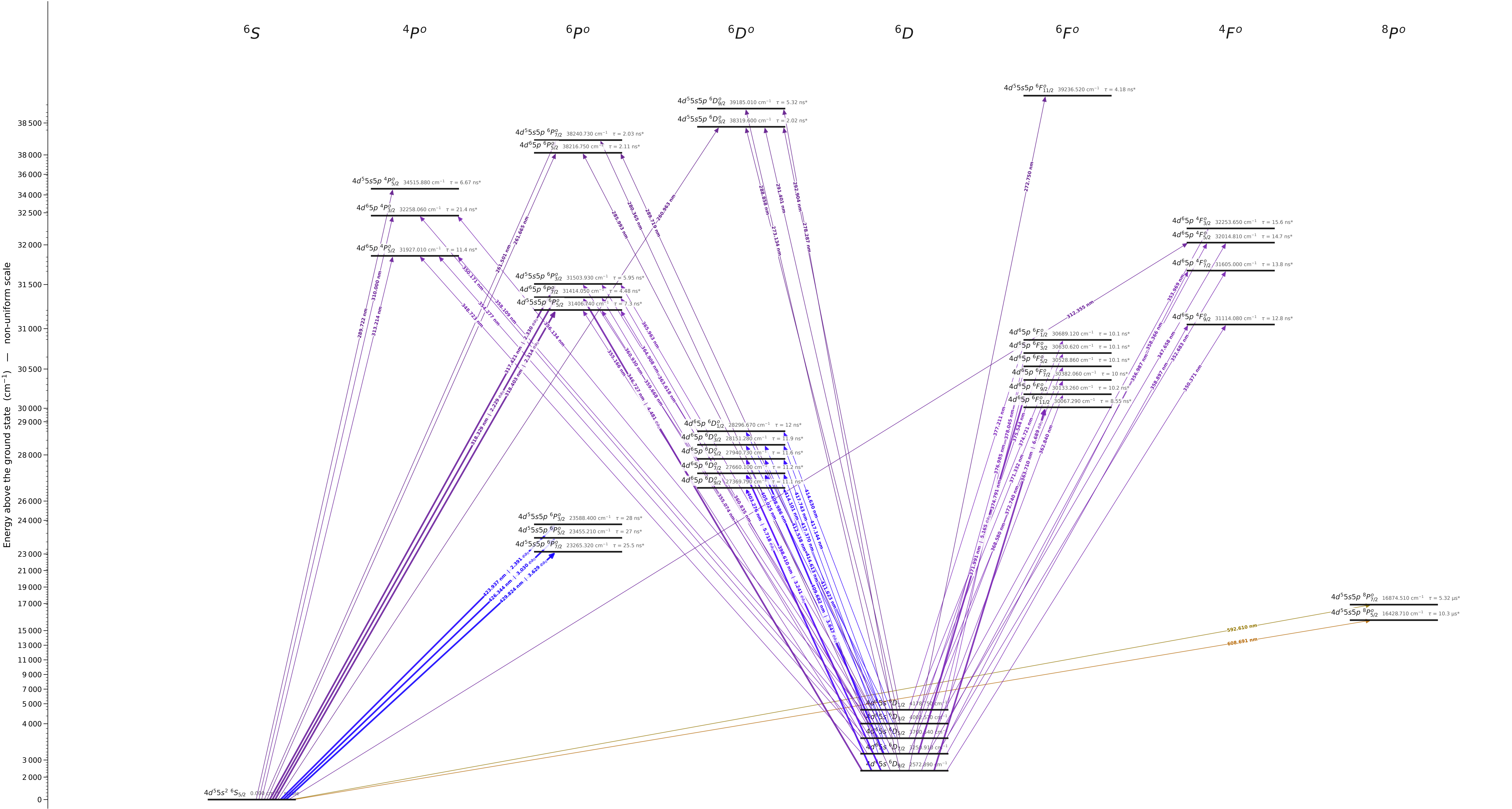
Technetium

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||J\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Tc (Tc I). Strongest classified transitions below 2 μm (NIST): 37 levels, 70 lines.

Tc⁺ ionisation limit 57421.68 cm⁻¹ = 7.11938 eV (174.150 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U er J\rangle $ (ea_0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4d^55s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{3/2}^o$	423.9372	423.8178	707.1624	2.391	3.800e+07	5.682	100	first step of the three-step Ti:sapphire resonance-ioniza...
$4d^55s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{5/2}^o$	426.3445	426.2245	703.1695	3.030	4.000e+07	5.889	100	first step of the three-step Ti:sapphire resonance-ioniza...
$4d^55s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{7/2}^o$	429.8243	429.7034	697.4768	3.629	4.200e+07	6.239	100	first step of the three-step Ti:sapphire resonance-ioniza...

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$4d^55s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{3/2}^o$	423.937	97-99	308(14)
$4d^55s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{5/2}^o$	423.937	98-99	324(14)
$4d^55s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{7/2}^o$	426.344	97-99	313(14)
$4d^55s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{3/2}^o$	426.344	98-99	327(14)
$4d^55s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{5/2}^o$	429.824	97-99	347(14)
$4d^55s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{7/2}^o$	429.824	98-99	369(14)

Shift = ν (first isotope) - ν (second isotope).

⁹⁸Tc hyperfine structure (I = 6)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$4d^55s^2\ ^6S_{5/2}$	-65.86(10)	2(6)	F=7/2: +1153.15 F=9/2: +856.32 F=11/2: +493.71 F=13/2: +65.44 F=15/2: -428.35 F=17/2: -987.52
$4d^55s5p\ ^6P_{3/2}^o$	491.0(3)	17(11)	F=5/2: -12021.76 F=7/2: -10306.30 F=9/2: -8100.04 F=11/2: -5402.49 F=13/2: -2213.05 F=15/2: +1469.02 F=17/2: +5644.52 F=19/2: +10314.41
$4d^55s5p\ ^6P_{5/2}^o$	548.6(3)	-9(11)	F=7/2: -9604.24 F=9/2: -7132.65 F=11/2: -4113.01 F=13/2: -545.99 F=15/2: +3567.55 F=17/2: +8226.65
$4d^55s5p\ ^6P_{7/2}^o$	726.9(3)	-3(10)	F=9/2: -7633.16 F=11/2: -3633.55 F=13/2: +1091.44 F=15/2: +6540.88

Measured A, B constants; sources in the data file.